

# DR. PRADEEP GUPTA

## Ph.D. in Computer Science & Engineering

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🎓 17+ Years Experience | 📖 40+ Publications | 🏆 6 SCIE and ESCI Papers Indexed | 📊 40+ Scopus

### 👤 PROFESSIONAL SUMMARY

Computer Scientist with a Ph.D. and 17+ years of academic, research, and administrative experience in Computer Science & Engineering. Expertise in **Artificial Intelligence, Computer Vision, Deep Learning, Federated Learning, and IoT-enabled systems**. Developed advanced hybrid models integrating **Vision Transformers, CNNs, GoogLeNet, and GANs** for applications in **smart agriculture and healthcare**. Demonstrated contributions in **SCI/Scopus publications, research supervision, funded projects, and academic leadership**.

Authored **40+ research publications**, including papers in **SCIE-indexed journals**, Scopus-indexed journals, book chapters, and international conference proceedings published by reputed venues such as Springer and JoVE. Demonstrates strong academic leadership through roles in **NBA coordination, NAAC support, ISO audit processes, departmental administration, conference organization, and student research mentoring**. Brings a mature university-ready profile combining research productivity, institutional service, and the capacity to build funded interdisciplinary research programmes.

### 🔍 RESEARCH IMPACT AND FUNDING POTENTIAL

- Developed original research frameworks including **PLANTDetectNet, PlantVitGNet, and Apple-ViT** for AI-enabled crop disease diagnosis and intelligent agricultural systems.
- Published in **SCIE-indexed journals** and Springer proceedings with interdisciplinary relevance spanning agriculture, computer vision, multimedia security, and intelligent healthcare.
- Active research directions align with national and international funding priorities in:
  - Artificial Intelligence for Smart Agriculture
  - Computer Vision and Explainable Deep Learning
  - Federated Learning and Privacy-Preserving AI
  - Edge AI and IoT-enabled Decision Systems
- Strong potential to contribute to funded projects under **DST, SERB, AICTE, ICAR, MeitY**, and related international collaborative programmes.

### 🏛️ EDUCATION

#### Postdoctoral Research Fellow

May 2026 – Jul 2027

National Kaohsiung University of Science and Technology (NKUST), Taiwan

**Laboratory:** Medical Physics and Informatics Laboratory  
**Supervisor:** Prof. Tsair-Fwu Lee  
**Research Topic:** Reliable and Explainable deep learning framework for Diabetic Retinopathy Grading  
**Key Responsibilities:** Designing deep learning-based image classification pipelines and preparing SCI-indexed manuscripts

## Ph.D. in Computer Science & Engineering

Awarded: 2026

Madhav Institute of Technology & Science, Deemed University Gwalior

- Dissertation:** Plant Disease Identification using Deep Learning Approaches
- Supervisor:** Prof. Rakesh Singh Jadon
- Key Contributions:** Proposed PLANTDetectNet, PlantVitGNet, Hybrid EFF-ViT Noisy and Apple-ViT for secure, robust, and deployable crop disease identification using hybrid deep learning models.
- Core Areas:** Vision Transformers, Deep Learning, Federated Learning, GAN-based Architectures, IoT-integrated Smart Agriculture

## M.Tech. in Computer Science & Engineering

2011

Institute of Engineering Technology, Bhagwant University, Ajmer

CGPA: 7.6/10

## B.E. in Computer Science & Engineering

2006

Madhav Institute of Technology & Science, Gwalior

RGPV Bhopal | 67.75%

## RESEARCH INTERESTS

Artificial Intelligence

Deep Learning

Computer Vision

Vision Transformers

Federated Learning

Smart Agriculture

IoT-enabled Systems

GANs

Explainable AI

## ACADEMIC APPOINTMENTS

### Assistant Professor, Department of Computer Science & Engineering

Aug 2019 – Present

Ajay Kumar Garg Engineering College (AKGEC), Ghaziabad

AKTU Affiliated

- Serving in teaching, research, accreditation, and academic administration roles in a leading engineering institution.
- **NBA Coordinator:** Led outcome-based education documentation and departmental accreditation processes.
- **NAAC Team Member:** Contributed to institutional quality assurance, compliance, and documentation.
- Served as **ISO Internal Audit In-Charge** for the department.
- Supervised **20+ B.Tech. capstone projects** in deep learning, computer vision, NLP, and IoT.

### Assistant Professor and Head of Department, CSE

Nov 2018 – Jul 2019

Gwalior Engineering College, Gwalior

RGPV Affiliated

- Headed the department and managed curriculum planning, academic delivery, faculty coordination, and student progression.
- Served concurrently as **Examination Superintendent**.

### Assistant Professor and Head of Department, CSE

Jul 2016 – Nov 2018

MITRC, Alwar, Rajasthan

RTU Affiliated

- Led the department with responsibility for curriculum execution, faculty mentoring, and laboratory development.
- Coordinated academic events and research activities at departmental level.
- Served as **Coordinator**, International Conference ETSEM-2018.
- Served as **Co-Convener**, National Conference ACCEE-2017.

### Assistant Professor and Head of Department, CSE & IT

Jul 2011 – Jul 2016

Gwalior Institute of Information Technology (GIIT), Gwalior

RGPV Affiliated

- Managed dual departments with responsibilities including academic planning, examination administration, budgeting, and laboratory infrastructure.
- Served as **Organising Secretary**, International Conference IRESM-2016.

## Assistant Professor, Department of Computer Science & Engineering

Jul 2006 – Jul 2009

Gwalior Engineering College, Gwalior

RGPV Affiliated

- Contributed to academic delivery, timetable planning, laboratory coordination, and examination logistics.

## PUBLICATIONS

Total Publications: **40+** | SCIE AND ESCI: **6** | Scopus Indexed: **40+**

### SCIE-Indexed Journal Articles

1. **P. Gupta**, S. Gupta, L. Goel, A. K. Agarwal, A. Singh, V. S. Sharma, C. L. Chowdhary, and R. Chowdhary, "PCE-FL: A Personalized, Clustered, and Communication-Efficient Federated Learning Framework for Robust Tomato Leaf Disease Detection," *AgriEngineering*, vol. 8, p. 182, 2026.
2. **P. Gupta** and R. S. Jadon, "PLANT Detect Net: A Hybrid IoT and Deep Learning Framework for Secure Plant Disease Detection and Classification," *Evolving Systems*, vol. 16, no. 2, p. 55, 2025.
3. **P. Gupta** and R. S. Jadon, "PlantVitGnet: A Hybrid Model of Vision Transformer and GoogLeNet for Plant Disease Identification," *Journal of Phytopathology*, vol. 173, no. 2, e70041, 2025.
4. **P. Gupta**, S. Gupta, L. Goel, A. K. Agarwal, A. Singh, and G. Ramesh, "Hybrid CNN and GAN Framework for Robust Deepfake Detection," *Journal of Visualized Experiments (JoVE)*, no. 222, 2025.
5. L. Goel, S. Gupta, A. Gupta, S. N. Rajan, V. K. Gupta, A. Singh, and **P. Gupta**, "Advancing ASD detection: novel approach integrating attention graph neural networks and crossover boosted meerkat optimization," *International Journal of Machine Learning and Cybernetics*, vol. 15, no. 8, pp. 3279–3297, Aug. 2024.
6. A. Jyoti, **P. Gupta**, S. Gupta, H. Khatter, and A. Mishra, "Inherent Insights Using Systematic Analytics of Development Tools in Ethereum Blockchain Smart Contracts," *Recent Advances in Electrical & Electronic Engineering*, vol. 18, no. 2, pp. 135–146, 2025.

### Scopus-Indexed Journal Articles and Book Chapters

7. **P. Gupta** and R. S. Jadon, "Noise-Resilient Hybrid EfficientNet–Vision Transformer Framework with Adaptive Symmetric Cross-Entropy Loss for Robust Plant Disease Detection," *Journal of Applied Science and Technology Trends*, vol. 7, no. 1, pp. 209–222, 2026.
8. **P. Gupta** and R. S. Jadon, "AppleVit: A Smart Agricultural Software for Apple Leaf Disease Detection Using AI," *Journal of Applied Science and Technology Trends*, vol. 6, no. 2, pp. 219–230, 2025.
9. S. Gupta, B. Verma, **P. Gupta**, L. Goel, A. K. Yadav, and D. Yadav, "Identification of Fake News Using Deep Neural Network-Based Hybrid Model," *SN Computer Science*, 2024.
10. **P. Gupta** and R. S. Jadon, "Comprehensive Study of Plant Disease Detection Using DL and ML Techniques," Book Chapter in *Research Handbook on Information Systems and Society*, Edward Elgar, accepted.

### Selected Scopus-Indexed Conference Publications

11. S. P. S. Kushwah, **P. Gupta**, and S. Gupta, "MammoCheck: Intelligent Mammogram Analysis for Breast Cancer Diagnosis Using CNNs and XGBoost," *Proceedings of ICDAM 2025, LNNS* vol. 1615, Springer, 2026.

12. S. Pandey, S. Gupta, **P. Gupta**, and A. Verma, "Quantum-Inspired Neural Networks, Federated Learning with Differential Privacy, and Hyperdimensional Computing for Enhanced Heart Disease Prediction," *Proceedings of ICDAM 2025*, LNNS vol. 1597, Springer, 2026.
13. Harshit et al. and **P. Gupta**, "Advanced Brain Imaging Analysis for Tumor Detection Using ML and DL," *Proceedings of ICICC 2025*, LNNS vol. 1434, Springer, 2025.
14. M. Atif et al. and **P. Gupta**, "Audio Sentiment Analysis Using Dataset Augmentation," *Proceedings of ICICC 2025*, LNNS vol. 1434, Springer, 2025.
15. S. Pandey, S. Gupta, **P. Gupta**, and A. Verma, "Heart Disease Risk Prediction Using Supervised ML Algorithms," *Proceedings of ICICC 2025*, LNNS vol. 1438, Springer, 2025.
16. G. Gautam et al. and **P. Gupta**, "Potato Disease Detection Using Machine Learning," *Proceedings of ICCCN 2024*, LNNS vol. 1294, Springer, 2025.
17. U. Sharma et al. and **P. Gupta**, "Automatic Bird Species Recognition Through Advanced ML Techniques," *Proceedings of ICCCN 2024*, LNNS vol. 1294, Springer, 2025.
18. P. Nagar et al. and **P. Gupta**, "Advancements in Plant Disease Detection Using Vision Transformers," *Proceedings of ICCCN 2024*, LNNS vol. 1294, Springer, 2025.
19. S. Pandey et al. and **P. Gupta**, "Stock Price Prediction Using LSTM," *Proceedings of ICDAM 2024*, LNNS vol. 1297, Springer, 2025.
20. A. Tyagi et al. and **P. Gupta**, "Advancements in Cartoonification Techniques: GAN-Based Algorithms," *AIP Conference Proceedings*, vol. 3224, 2024.
21. **P. Gupta**, S. Gupta, L. Goel, V. Yadav, and D. Singh, "Hate Speech Detection: Recent Advancements and Emerging Technologies," *Proceedings of ICICC 2024*, LNNS vol. 1043, Springer, 2024.
22. D. Singh, S. Gupta, and **P. Gupta**, "Detection of Hate Speech Using Ensemble Models," *Proceedings of ICICC 2024*, LNNS vol. 1021, Springer, 2024.
23. D. Choudhary, **P. Gupta**, and S. Gupta, "Diabetes Prediction Using ML Classifiers," *Proceedings of MIND 2023*, CCIS vol. 2128, Springer, 2023.
24. D. Choudhary, **P. Gupta**, and S. Gupta, "Early Phase Detection of Diabetes Mellitus Using ML," *Proceedings of IACC 2023*, Springer, 2023.
25. Deepak, S. Gupta, and **P. Gupta**, "Plant Disease Recognition Using ML and DL Classifiers," *Proceedings of IACC 2023*, Springer, 2023.
26. D. Singh et al. and **P. Gupta**, "A Novel Approach to Image Restoration and Enhancement," *Proceedings of ICCCN 2023*, LNNS vol. 917, Springer, 2023.
27. S. Singh et al. and **P. Gupta**, "Handwritten Digit Recognition Using ML Classifier," *Proceedings of ICCCN 2023*, LNNS vol. 977, Springer, 2023.
28. L. Mahur, S. Gupta, and **P. Gupta**, "ML Techniques to Detect Depression from Social Media," *Proceedings of ICACITE 2023*, 2023.
29. **P. Gupta** and R. S. Jadon, "Plant Disease Detection Using Machine Learning Models," *Proceedings of ICCS 2023*, LPU, India, 2023.
30. R. Abhishek et al. and **P. Gupta**, "Mood-Based Music Recommendation System Using Supervised Learning," *Proceedings of DOSCI 2023*, 2023.
31. L. Mahur, S. Gupta, and **P. Gupta**, "Review on Depression Detection on Social Media Using ML," *Proceedings of DOSCI 2023*, 2023.
32. G. Singh, C. Agarwal, I. Kaur, and **P. Gupta**, "ML for Cardiac Arrhythmia Detection: A Systematic Survey," *ICAKGEC 2023*, 2023.

## Earlier Publications

33. N. K. Gupta, N. P. Gupta, **P. Gupta**, and K. Kumar, "Error-Controlling Technique in Wireless Communication," *Proceedings of ICSEC 2021*, AIS, Springer, 2021.
34. **P. Gupta** and R. S. Pippal, "Secure Framework for Administrator Authentication in UTM

- System," *National Journal of Multidisciplinary Research and Development*, vol. 3, no. 1, 2018.
35. R. S. Pippal, **P. Gupta**, and R. Singh, "A Novel Smart Card Authentication Scheme Using Image Encryption," *International Journal of Computer Applications*, vol. 72, no. 9, pp. 8–14, 2013.
36. **P. Gupta** and S. Garg, "Authentication Scheme of Smart Card by Image Watermarking," *Proceedings of ACCEE 2017, MITRC Alwar*, 2017.
37. **P. Gupta** and S. Garg, "A Novel Smart Card Authentication Scheme Using Image Watermarking," *Proceedings of IRESM 2016, GIIT Gwalior*, 2016.
38. **P. Gupta** and S. P. S. Kushwah, "Localisation for Ad-Hoc Sensor Networks," *Proceedings of National Conference on Information Technology, ITM University*, 2011.

## DOCTORAL RESEARCH CONTRIBUTIONS

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- **PLANTDetectNet**: Proposed a hybrid IoT and deep learning framework for secure, real-time plant disease detection and classification using edge-aware intelligent processing.
- **PlantVitGNet**: Designed a novel hybrid fusion of Vision Transformer and GoogLeNet for improved plant disease identification, combining local and global feature representation.
- **Noisy Hybrid EffVit**: Developed a hybrid EfficientNet–ViT model with ASCE loss for noise-robust plant disease detection. Achieved improved accuracy and generalization under real-world noisy conditions.
- **AppleViT**: Developed an AI-driven software framework for automated apple leaf disease recognition with practical deployment relevance for precision agriculture.
- Extended methodological expertise to **deepfake detection** using hybrid CNN and GAN frameworks for robust multimedia forensics.
- Current forward-looking research includes **federated learning for privacy-preserving agricultural AI**, **edge-deployable transformer architectures**, and **multi-sensor intelligent disease monitoring systems**.

## RESEARCH PROJECTS AND GUIDED WORK

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- Research direction under development on **federated learning for smart crop disease detection** with potential application to distributed agricultural monitoring systems.
- Guided student research and applied projects in:
  - Breast cancer and mammogram analysis
  - Brain tumour detection
  - Hate speech and fake news detection
  - Deepfake detection
  - Depression detection from social media
  - Crop disease diagnosis using machine learning and deep learning
- Built a research culture resulting in **22 Scopus-indexed student-led publications**.

## ACADEMIC LEADERSHIP AND ACCREDITATION

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- **Head of Department** at three institutions across academic administration, curriculum planning, faculty mentoring, and departmental development.
- **NBA Coordinator** for outcome-based education documentation and accreditation support.
- **NAAC Accreditation Contributor** for institutional quality assurance and compliance activities.
- **ISO Internal Audit In-Charge** for departmental academic process standardization.
- **Examination Superintendent** and academic administration roles in multiple institutions.



## CONFERENCE ORGANISATION AND SERVICE

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- **Organising Secretary**, Ajay Kumar Garg Engineering College, successfully organized the **1st IEEE International Conference on Advanced Computing and Emerging Technologies (ACET-2024)**, promoting research and innovation in computing domains.
- **Organising Secretary**, Ajay Kumar Garg Engineering College, organized the **2nd IEEE International Conference on Advanced Computing and Emerging Technologies (ACET-2025)**, focusing on emerging trends in advanced computing.
- **Coordinator**, International Conference on Emerging Trends in Science, Engineering & Management (ETSEM-2018), MITRC Alwar, 2018.
- **Co-Convener**, 3rd National Conference on Advances in Communication, Computer & Electrical Engineering (ACCCEE-2017), MITRC Alwar, 2017.
- **Organising Secretary**, International Conference on Innovative Research on Engineering Science & Management (IRESM-2016), GIIT Gwalior, 2016.
- **Organising Committee Member**, FDP on Advancement of AI in Cyber Security, AKGEC Ghaziabad, 2023.
- Contributed to workshops and faculty development activities in robotics, ethical hacking, Cisco technology, Microsoft outreach programmes, and Wipro Mission-10X initiatives.

## TEACHING PORTFOLIO

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*Representative undergraduate courses taught across institutions:*

Machine Learning

Theory of Automata & Formal Languages

Compiler Design

Human-Computer Interaction

Project Management

Entrepreneurship

## STUDENT MENTORSHIP AND SUPERVISION

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- Supervised and co-supervised **20+ B.Tech. capstone projects** in AI, deep learning, NLP, computer vision, and IoT-based applications.
- Mentored student research leading to **22 Scopus-indexed publications** between 2023 and 2026.
- Guided projects in healthcare AI, social media analytics, multimedia forensics, recommendation systems, and agricultural intelligence.
- Supported students in publishing and presenting work at national and international conferences.

## PROFESSIONAL DEVELOPMENT AND TRAINING

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- **AICTE ATAL FDP on Artificial Intelligence (AI) Leading to Academic Excellence: Transforming Academic Writing & Research**  
Organization: Oriental College of Management, Bhopal  
Duration: 12–17 January 2026.
- **AICTE ATAL FDP on Bridging Theory to Practice: High Performance Computing, Artificial Intelligence, and Data-Driven Solutions**  
Organization: BML Munjal University  
Duration: 06–11 October 2025.
- **AICTE ATAL FDP on Cyber Security & Applications**  
Organization: Vellore Institute of Technology, Andhra Pradesh  
Duration: 01–06 September 2025.
- **AICTE ATAL FDP on Quantum Computing with Artificial Intelligence: Convergence and**

### Applications

Organization: Gayatri Vidya Parishad College for Degree and PG Courses

Duration: 25–30 August 2025.

➤ **AICTE ATAL FDP on Smart Healthcare Systems: Enhancing Patient Care with IoT and AI**

Organization: Amity School of Engineering & Technology, Amity University

Duration: 02–07 December 2024.

➤ **FDP on Entrepreneurship, Incubation, and Innovation**

Organization: Department of Biotechnology, Bansal Institute of Engineering and Technology, Lucknow

Duration: 5–10 August 2024.

➤ **FDP on Smart Healthcare Systems: Enhancing Patient Care with IoT and AI (ATAL)**

Organization: Amity School of Engineering & Technology, Amity University

Duration: 2–7 December 2024.

➤ **FDP on Next Generation Technology of Information and Communication**

Organization: Bengal College of Information & Technology

Duration: 5–9 November 2024.

## TECHNICAL PROFICIENCIES

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|-------------------------------------|----------------------------------------------------------------------------------|
| <b>Programming Languages</b>        | Python, MATLAB, C, C++, Java                                                     |
| <b>Frameworks and Libraries</b>     | TensorFlow, PyTorch, Keras, Scikit-learn                                         |
| <b>Research and Authoring Tools</b> | L <sup>A</sup> T <sub>E</sub> X, Overleaf, Jupyter Notebook, Google Colab        |
| <b>Domains of Expertise</b>         | Deep Learning, Computer Vision, NLP, IoT, Federated Learning, GANs, Transformers |